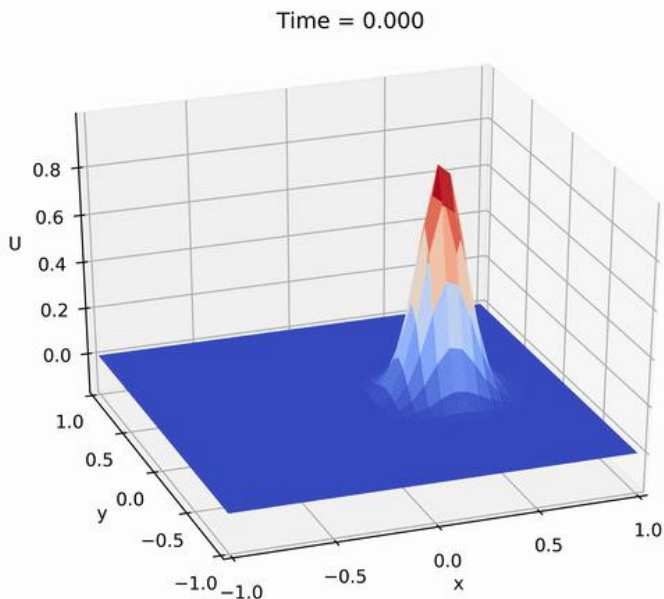




AIMS

Numerical Analysis in Python



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Numerical Differentiation



- First order derivatives
- High order derivatives
- Examples

Motivation

- How do you evaluate the derivative of a tabulated function.
- How do we determine the velocity and acceleration from tabulated measurements.

Time (second)	Displacement (meters)
0	30.1
5	48.2
10	50.0
15	40.2

Recall

$$\frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Taylor Theorem :

$$f(x+h) = f(x) + f'(x)h + \frac{f^{(2)}(x)h^2}{2!} + \frac{f^{(3)}(x)h^3}{3!} + O(h^4)$$

$$E = O(h^n) \Rightarrow \exists \text{ real, finite } C, \text{ such that: } |E| \leq C|h|^n$$

E is of order $h^n \Rightarrow E$ is approaching zero at rate similar to h^n

Chapter 5: Numerical Differentiation

$$f'(x_0) = \frac{f(x_0 + h) - f(x_0)}{h} - \frac{h}{2} f''(\xi).$$

For small values of h , the difference quotient $[f(x_0 + h) - f(x_0)]/h$ can be used to approximate $f'(x_0)$ with an error bounded by $M|h|/2$, where M is a bound on $|f''(x)|$ for x between x_0 and $x_0 + h$.

Three Formula

Forward Difference : $\frac{df(x)}{dx} = \frac{f(x+h) - f(x)}{h}$

Backward Difference : $\frac{df(x)}{dx} = \frac{f(x) - f(x-h)}{h}$

Central Difference : $\frac{df(x)}{dx} = \frac{f(x+h) - f(x-h)}{2h}$

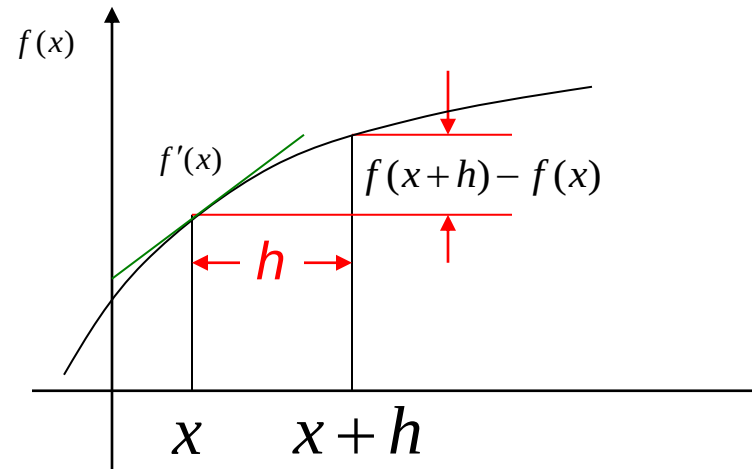
Which method is better? How do we judge them?

Chapter 5: Numerical Differentiation

Forward Difference Formula for $f'(x)$

$$f'(x) \approx \frac{f(x+h) - f(x)}{h} \quad \text{error} = O(h)$$

Geometrically



Backward Difference Formula for $f'(x)$

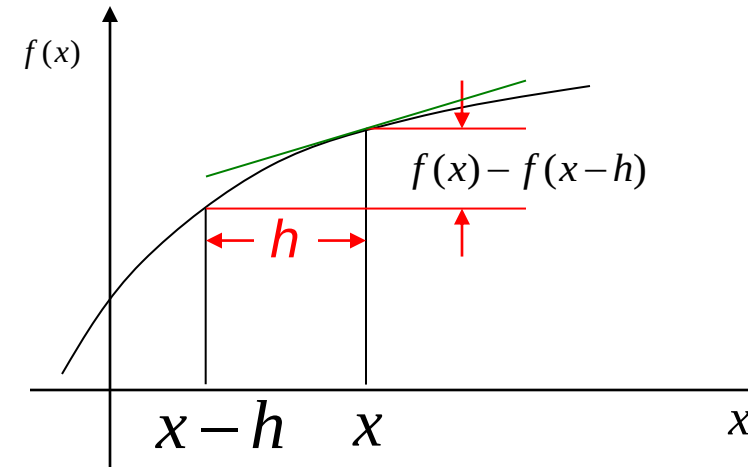
Similarly

$$f(x-h) = f(x) - f'(x)h + \frac{1}{2} f''(x)h^2 + O(h^3)$$

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

$error = O(h)$

Geometrically



Forward/Backward Difference Formula

Forward Difference :

$$\begin{aligned}f(x+h) &= f(x) + f'(x)h + O(h^2) \\ \Rightarrow f'(x)h &= f(x+h) - f(x) + O(h^2) \\ \Rightarrow f'(x) &= \frac{f(x+h) - f(x)}{h} + O(h)\end{aligned}$$

Backward Difference :

$$\begin{aligned}f(x-h) &= f(x) - f'(x)h + O(h^2) \\ \Rightarrow f'(x)h &= f(x) - f(x-h) + O(h^2) \\ \Rightarrow f'(x) &= \frac{f(x) - f(x-h)}{h} + O(h)\end{aligned}$$

Central Difference Formula

Central Difference :

$$f(x+h) = f(x) + f'(x)h + \frac{f^{(2)}(x)h^2}{2!} + \frac{f^{(3)}(x)h^3}{3!} + \frac{f^{(4)}(x)h^4}{4!} + \dots$$

$$f(x-h) = f(x) - f'(x)h + \frac{f^{(2)}(x)h^2}{2!} - \frac{f^{(3)}(x)h^3}{3!} + \frac{f^{(4)}(x)h^4}{4!} + \dots$$

$$f(x+h) - f(x-h) = 2f'(x)h + 2\frac{f^{(3)}(x)h^3}{3!} + \dots$$

$$\Rightarrow f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O(h^2)$$

The Three Formula (Revisited)

Forward Difference :
$$\frac{df(x)}{dx} = \frac{f(x+h) - f(x)}{h} + O(h)$$

Backward Difference :
$$\frac{df(x)}{dx} = \frac{f(x) - f(x-h)}{h} + O(h)$$

Central Difference :
$$\frac{df(x)}{dx} = \frac{f(x+h) - f(x-h)}{2h} + O(h^2)$$

Forward and backward difference formulas are comparable in accuracy.
Central difference formula is expected to give a better answer.

Higher Order Formulas

$$f(x+h) = f(x) + f'(x)h + \frac{f^{(2)}(x)h^2}{2!} + \frac{f^{(3)}(x)h^3}{3!} + \frac{f^{(4)}(x)h^4}{4!} + \dots$$

$$f(x-h) = f(x) - f'(x)h + \frac{f^{(2)}(x)h^2}{2!} - \frac{f^{(3)}(x)h^3}{3!} + \frac{f^{(4)}(x)h^4}{4!} + \dots$$

$$f(x+h) + f(x-h) = 2f(x) + 2\frac{f^{(2)}(x)h^2}{2!} + 2\frac{f^{(4)}(x)h^4}{4!} + \dots$$

$$\Rightarrow f^{(2)}(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2)$$


$$\text{Error} = -\frac{f^{(4)}(\xi)h^2}{12}$$

Other Higher Order Formulas

$$f^{(2)}(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$$

$$f^{(3)}(x) = \frac{f(x+2h) - 2f(x+h) + 2f(x-h) - f(x-2h)}{2h^3}$$

$$f^{(4)}(x) = \frac{f(x+2h) - 4f(x+h) + 6f(x) - 4f(x-h) + f(x-2h)}{h^4}$$

Central Formulas with $Error = O(h^2)$

Other formulas for $f^{(2)}(x)$, $f^{(3)}(x)$... are also possible.

You can use Taylor Theorem to prove them and obtain the error order.

Example

- Use forward, backward and centered difference approximations to estimate the first derivative of:

$$f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$$

at $x = 0.5$ using step size $h = 0.5$ and $h = 0.25$

- Note that the derivative can be obtained directly:

$$f'(x) = -0.4x^3 - 0.45x^2 - 1.0x - 0.25$$

The true value of $f'(0.5) = -0.9125$

- In this example, the function and its derivative are known. However, in general, only tabulated data might be given.

Solution with Step Size = 0.5

- $f(0.5) = 0.925$, $f(0) = 1.2$, $f(1.0) = 0.2$
- Forward Divided Difference:
 $f'(0.5) \approx (0.2 - 0.925)/0.5 = -1.45$
 $|\varepsilon_t| = |(-0.9125 + 1.45)/-0.9125| = 58.9\%$
- Backward Divided Difference:
 $f'(0.5) \approx (0.925 - 1.2)/0.5 = -0.55$
 $|\varepsilon_t| = |(-0.9125 + 0.55)/-0.9125| = 39.7\%$
- Centered Divided Difference:
 $f'(0.5) \approx (0.2 - 1.2)/1.0 = -1.0$
 $|\varepsilon_t| = |(-0.9125 + 1.0)/-0.9125| = 9.6\%$

Solution with Step Size = 0.25

- $f(0.5)=0.925$, $f(0.25)=1.1035$, $f(0.75)=0.6363$
- Forward Divided Difference:
$$f'(0.5) \approx (0.6363 - 0.925)/0.25 = -1.155$$
$$|\varepsilon_t| = |(-0.9125 + 1.155)/-0.9125| = 26.5\%$$
- Backward Divided Difference:
$$f'(0.5) \approx (0.925 - 1.1035)/0.25 = -0.714$$
$$|\varepsilon_t| = |(-0.9125 + 0.714)/-0.9125| = 21.7\%$$
- Centered Divided Difference:
$$f'(0.5) \approx (0.6363 - 1.1035)/0.5 = -0.934$$
$$|\varepsilon_t| = |(-0.9125 + 0.934)/-0.9125| = 2.4\%$$

Discussion

- For both the Forward and Backward difference, the error is $O(h)$
- Halving the step size h approximately halves the error of the Forward and Backward differences
- The Centered difference approximation is more accurate than the Forward and Backward differences because the error is $O(h^2)$
- Halving the step size h approximately quarters the error of the Centered difference.

Example 2

Use the forward-difference formula to approximate the derivative of $f(x) = \ln x$ at $x_0 = 1.8$ using $h = 0.1$, $h = 0.05$, and $h = 0.01$, and determine bounds for the approximation errors.

Solution

The forward-difference formula

$$\frac{f(1.8 + h) - f(1.8)}{h}$$

with $h = 0.1$ gives

$$(\ln 1.9 - \ln 1.8)/0.1$$

$$= (0.64185389 - 0.58778667)/0.1 = 0.5406722.$$

Because $f''(x) = -1/x^2$ and $1.8 < \xi < 1.9$, a bound for this approximation error is $|hf''(\xi)|/2$

$$= |h|/2\xi^2 < 0.1/2(1.8)^2 = 0.0154321.$$

Chapter 5: Numerical Differentiation

h	$f(1.8 + h)$	$\frac{f(1.8 + h) - f(1.8)}{h}$	$\frac{ h }{2(1.8)^2}$
0.1	0.64185389	0.5406722	0.0154321
0.05	0.61518564	0.5479795	0.0077160
0.01	0.59332685	0.5540180	0.0015432

Euler's Method

Leonhard Euler made a huge number of contributions to mathematics, almost half after he was totally blind.

(When this portrait was made he had already lost most of the sight in his right eye.)



Leonhard Euler 1707 - 1783

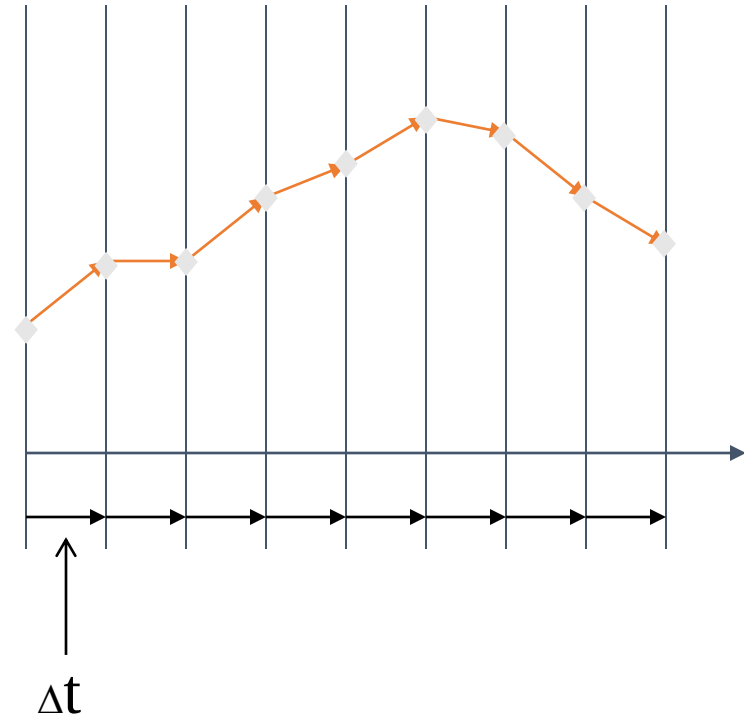


Euler's Method

Suppose we know the value of the derivative of a function at every point and we know the value of the function at one point.

We can build an approximate graph of the function using local linearity to approximate over and over again. This iterative procedure is called Euler's Method.

Here's how it works.



Implementing Euler's Method

What's needed to get Euler's method started?

- Well, you need a differential equation of the form:

$$y' = \text{some expression in } t \text{ and } y$$

- And a point (t_0, y_0) that lies on the graph of the solution function $y = f(t)$.

- Finally, you need a fixed step size

$$\Delta t.$$

A smaller step size will lead to more accuracy, but will also take more computations.

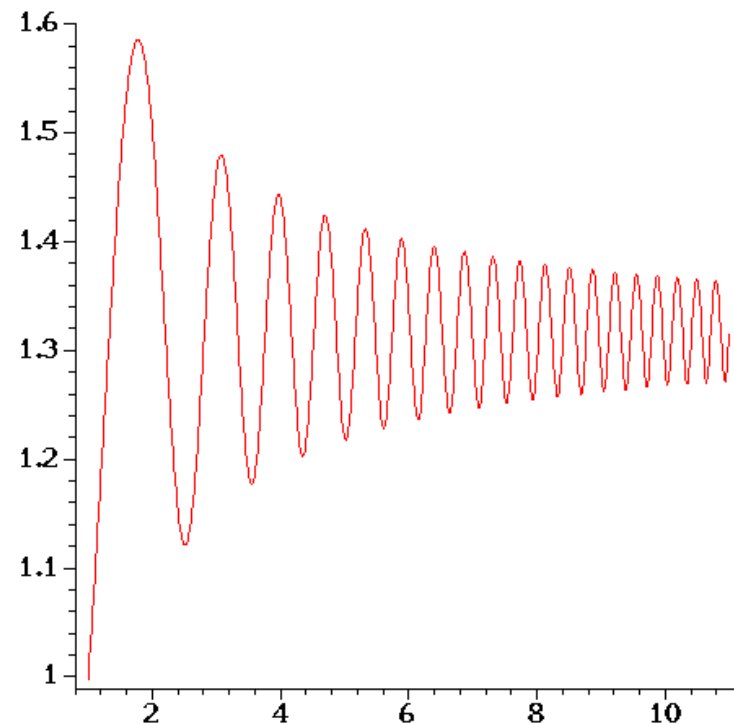
$$\text{New } t = \text{Old } t + \Delta t$$

$$\text{New } y = \text{Old } y + \Delta y = \text{Old } y + (\text{slope at (Old } t, \text{ Old } y)) \Delta t$$

For instance, if

$$y' = \sin(t^2)$$

and $(1,1)$ lies on the graph of $y = f(t)$, then 1000 steps of length .01 yield the following graph of the function f .



This graph is the anti-derivative of $\sin(t^2)$; a function which has no elementary formula!

How do we accomplish this?

Suppose that $y' = t \sin(y)$ and $(1,1)$ lies on the graph.

Let $\Delta t = .1$.

Old Point		Slope at old Pt.	Change in y	New t	New y
Old t	Old y	$y'(\text{old } t, \text{old } y)$	$Dy = y' * Dt$	Old t + Dt	Oldy + Dy
1	1	.8414	.08414	1.1	1.084
1.1	1.084	.9723	.09723	1.2	1.181
1.2	1.181	1.1101	.11101	1.3	1.292

$$\text{New } t = \text{Old } t + \Delta t$$

$$\text{New } y = \text{Old } y + \Delta y = \text{Old } y + y'(\text{Old } t, \text{Old } y) \Delta t$$

Euler's Method

$$\frac{dy}{dx} = f(x, y), y(0) = y_0$$

$$\begin{aligned} \text{Slope} &= \frac{\text{Rise}}{\text{Run}} \\ &= \frac{y_1 - y_0}{x_1 - x_0} \\ &= f(x_0, y_0) \end{aligned}$$

$$\begin{aligned} y_1 &= y_0 + f(x_0, y_0)(x_1 - x_0) \\ &= y_0 + f(x_0, y_0)h \end{aligned}$$

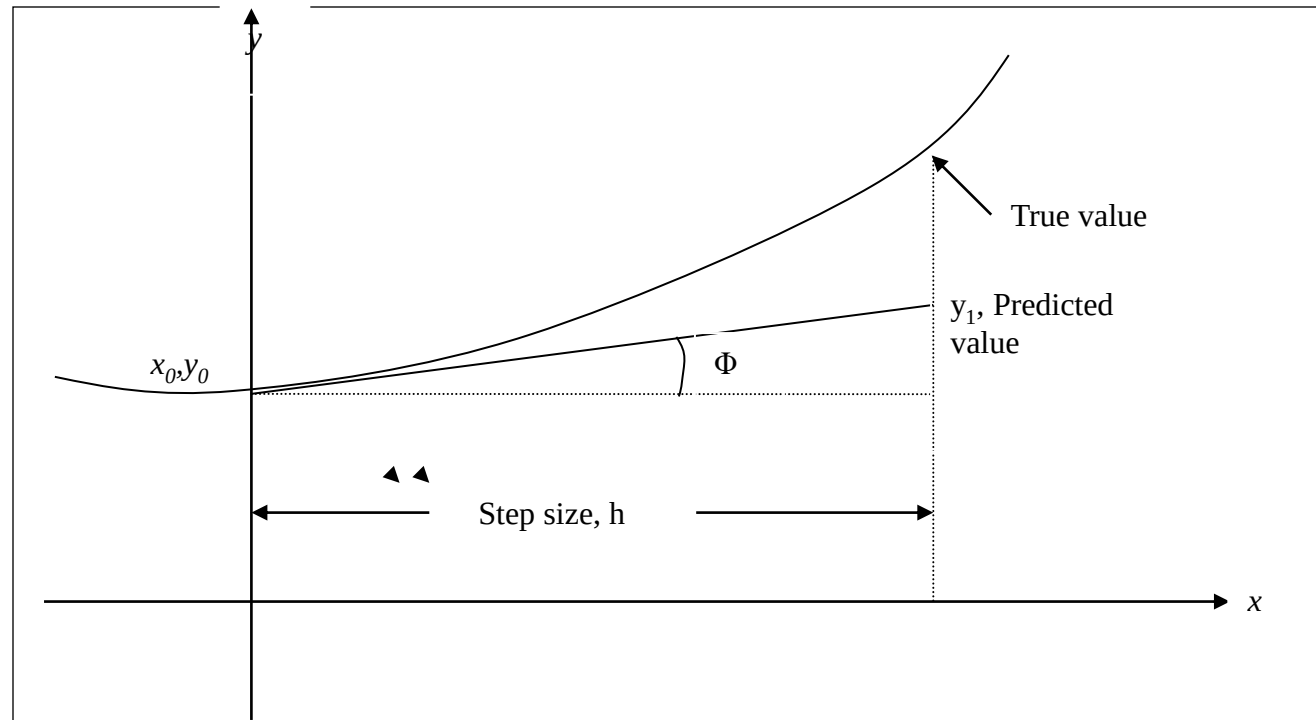


Figure 1 Graphical interpretation of the first step of Euler's method

Euler's Method

$$y_{i+1} = y_i + f(x_i, y_i)h$$

$$h = x_{i+1} - x_i$$

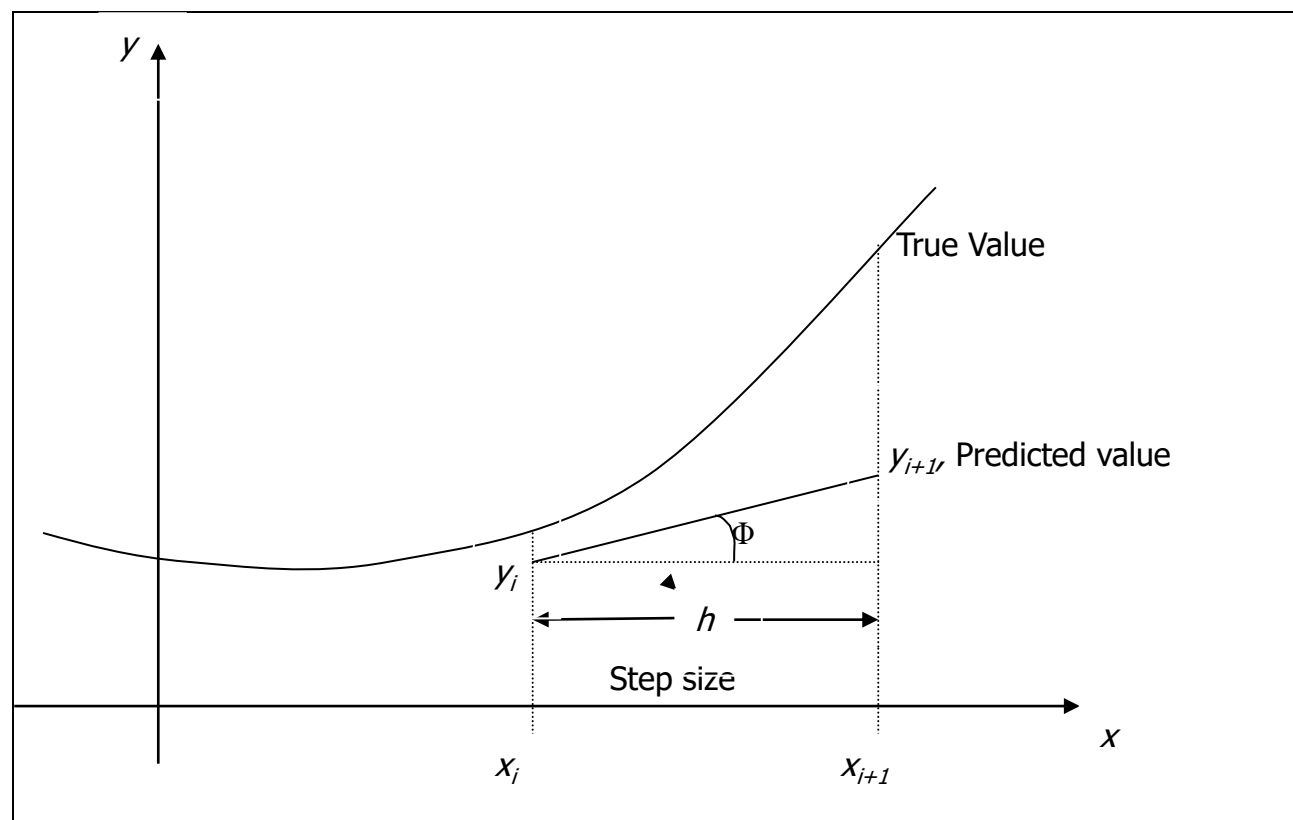


Figure 2. General graphical interpretation of Euler's method

How to write Ordinary Differential Equation

How does one write a first order differential equation in the form of

$$\frac{dy}{dx} = f(x, y)$$

Example

$$\frac{dy}{dx} + 2y = 1.3e^{-x}, y(0) = 5$$

is rewritten as

$$\frac{dy}{dx} = 1.3e^{-x} - 2y, y(0) = 5$$

In this case

$$f(x, y) = 1.3e^{-x} - 2y$$

Example

A ball at 1200K is allowed to cool down in air at an ambient temperature of 300K. Assuming heat is lost only due to radiation, the differential equation for the temperature of the ball is given by

$$\frac{d\theta}{dt} = -2.2067 \times 10^{-12} (\theta^4 - 81 \times 10^8), \theta(0) = 1200K$$

Find the temperature at $t = 480$ seconds using Euler's method. Assume a step size of $h = 240$ seconds.

Solution

Step 1:

$$\frac{d\theta}{dt} = -2.2067 \times 10^{-12} (\theta^4 - 81 \times 10^8)$$

$$f(t, \theta) = -2.2067 \times 10^{-12} (\theta^4 - 81 \times 10^8)$$

$$\theta_{i+1} = \theta_i + f(t_i, \theta_i)h$$

$$\theta_1 = \theta_0 + f(t_0, \theta_0)h$$

$$= 1200 + f(0, 1200)240$$

$$= 1200 + (-2.2067 \times 10^{-12} (1200^4 - 81 \times 10^8))240$$

$$= 1200 + (-4.5579)240$$

$$= 106.09K$$

θ_1 is the approximate temperature at $t = t_1 = t_0 + h = 0 + 240 = 240$

$$\theta(240) \approx \theta_1 = 106.09K$$

Solution Cont

Step 2: For $i = 1$, $t_1 = 240$, $\theta_1 = 106.09$

$$\begin{aligned}\theta_2 &= \theta_1 + f(t_1, \theta_1)h \\ &= 106.09 + f(240, 106.09)240 \\ &= 106.09 + \left(-2.2067 \times 10^{-12} (106.09^4 - 81 \times 10^8)\right)240 \\ &= 106.09 + (0.017595)240 \\ &= 110.32K\end{aligned}$$

θ_2 is the approximate temperature at $t = t_2 = t_1 + h = 240 + 240 = 480$

$$\theta(480) \approx \theta_2 = 110.32K$$

Solution Cont

The exact solution of the ordinary differential equation is given by the solution of a non-linear equation as

$$0.92593 \ln \frac{\theta - 300}{\theta + 300} - 1.8519 \tan^{-1}(0.00333\theta) = -0.22067 \times 10^{-3} t - 2.9282$$

The solution to this nonlinear equation at $t=480$ seconds is

$$\theta(480) = 647.57 K$$

Comparison of Exact and Numerical Solutions

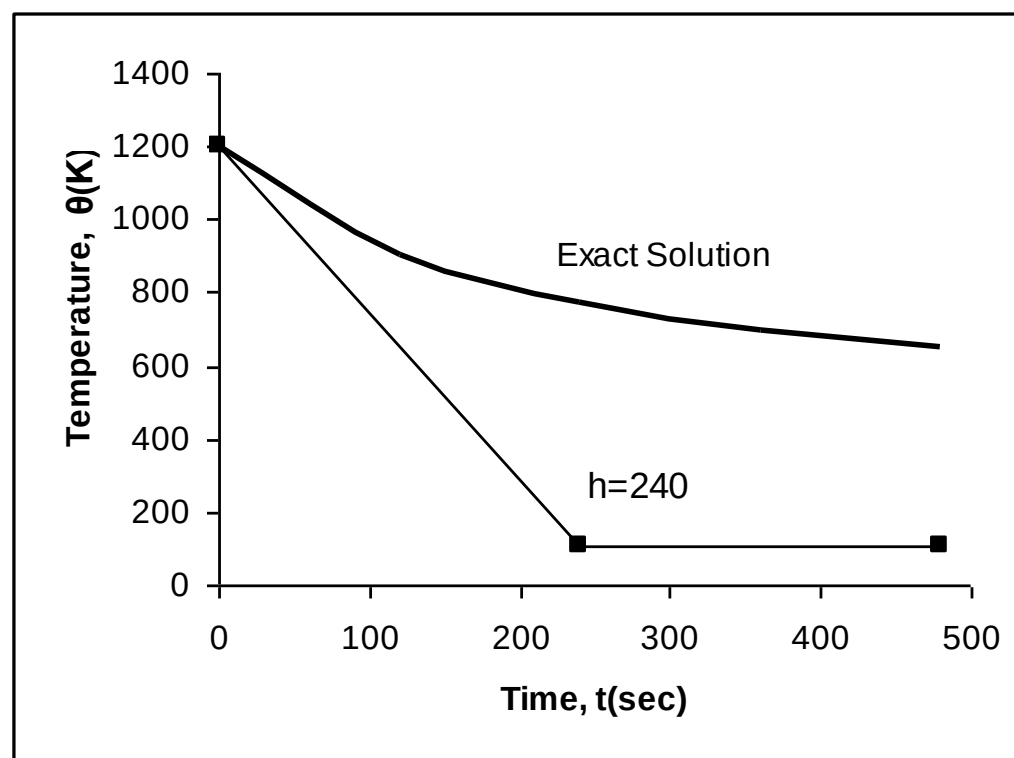


Figure 3. Comparing exact and Euler's method

Effect of step size

Table 1. Temperature at 480 seconds as a function of step size, h

Step, h	$\theta(480)$	E_t	$ \epsilon_t \%$
480	-987.81	1635.4	252.54
240	110.32	537.26	82.964
120	546.77	100.80	15.566
60	614.97	32.607	5.0352
30	632.77	14.806	2.2864

$$\theta(480) = 647.57K \quad (\text{exact})$$

Comparison with exact results

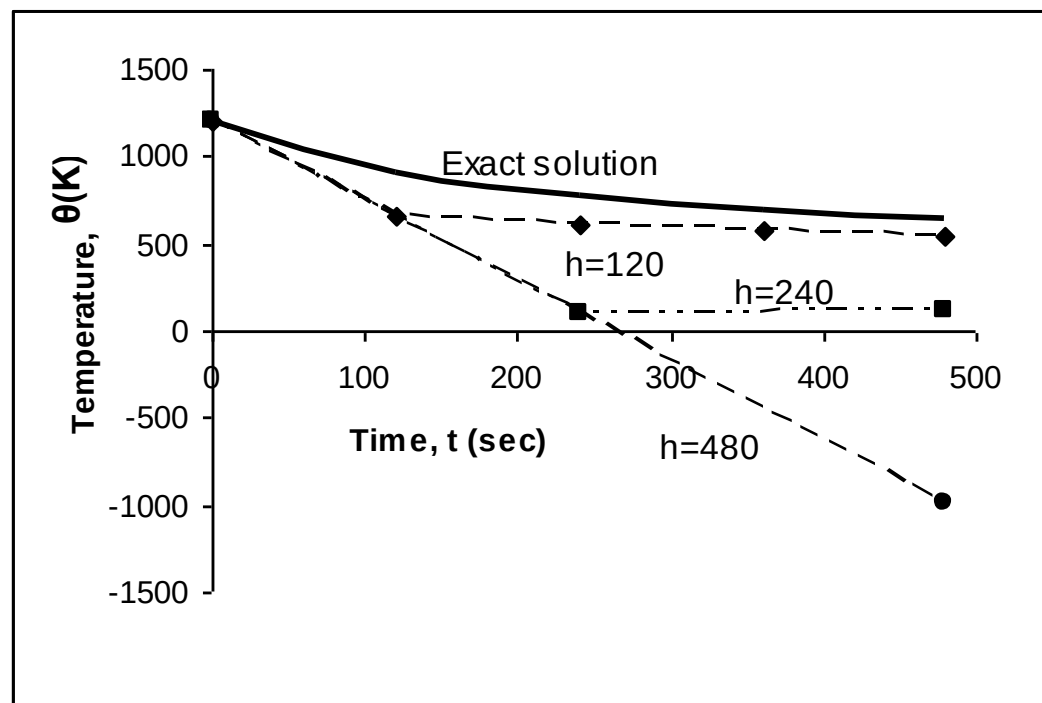


Figure 4. Comparison of Euler's method with exact solution for different step sizes

Effects of step size on Euler's Method

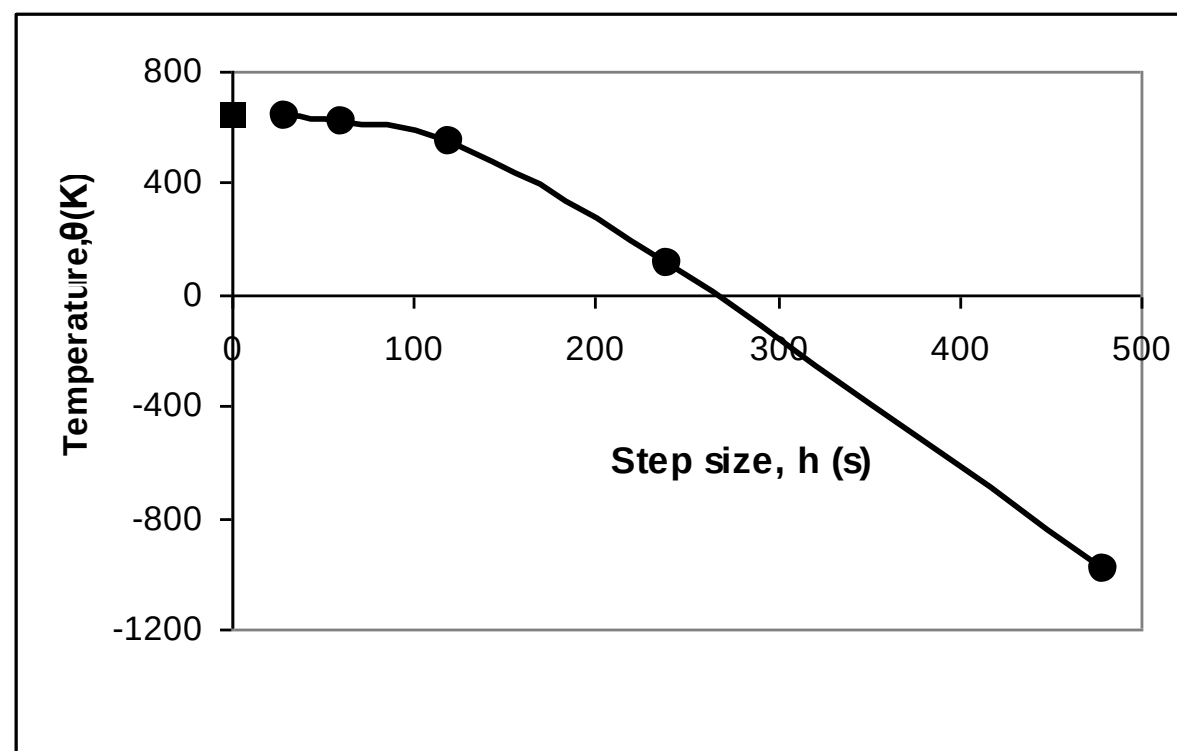


Figure 5. Effect of step size in Euler's method.

Errors in Euler's Method

It can be seen that Euler's method has large errors. This can be illustrated using Taylor series.

$$y_{i+1} = y_i + \left. \frac{dy}{dx} \right|_{x_i, y_i} (x_{i+1} - x_i) + \frac{1}{2!} \left. \frac{d^2 y}{dx^2} \right|_{x_i, y_i} (x_{i+1} - x_i)^2 + \frac{1}{3!} \left. \frac{d^3 y}{dx^3} \right|_{x_i, y_i} (x_{i+1} - x_i)^3 + \dots$$

$$y_{i+1} = y_i + f(x_i, y_i)(x_{i+1} - x_i) + \frac{1}{2!} f'(x_i, y_i)(x_{i+1} - x_i)^2 + \frac{1}{3!} f''(x_i, y_i)(x_{i+1} - x_i)^3 + \dots$$

As you can see the first two terms of the Taylor series

$$y_{i+1} = y_i + f(x_i, y_i)h \quad \text{are the Euler's method.}$$

The true error in the approximation is given by

$$E_t = \frac{f'(x_i, y_i)}{2!} h^2 + \frac{f''(x_i, y_i)}{3!} h^3 + \dots \quad E_t \propto h^2$$